

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } e_0 \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [↓base
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } e_1 \text{ } l) F \dots) FS_1 \dots))$
 where $(\text{not value?}[e]), (\sigma_1 e_1) = \Downarrow \text{base}[\sigma_0, e_0], t_1 = \text{step}$
 $(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{spawn (tag } x_{\text{label}})]) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [task-spaw
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] \text{ } l) F \dots) FS_1 \dots))$
 $(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{run } v)] \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [task-r
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] \text{ } l) F \dots) FS_1 \dots))$
 where $(\text{ptr } x_{\text{async}}) = \text{malloc}$
 $(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } v \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [task-retu
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$
 where $(\text{async? } l), x_{\text{async}} = l, v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{a}}$
 $(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame (throw } v) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [task-fail
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$
 where $(\text{async? } l), x_{\text{async}} = l, v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{a}}$
 $(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame$ $\longrightarrow$ [await-suspe
 $\quad E[(\text{tagged } x_{\text{running}} \text{ } J[(\text{await (task } x_{\text{async}})])]) \text{ } l)$
 $\quad F \dots) FS_1 \dots))$
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$
 $(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [await-contin
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } E[v] \text{ } l) F \dots) FS_1 \dots))$
 where $(\text{done } v) = \text{task-state}[\text{lookup}[\sigma_0, x_{\text{asy}}$
 $(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [await-fail
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } E[(\text{throw } v)] \text{ } l) F \dots) FS_1 \dots))$
 where $v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}], (\text{failed } v) = \text{task-state}[\text{lookup}[\sigma_0, x_{\text{asy}}$
 $(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{cancel (task } x_{\text{async}})]) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [cancel
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } E[(\text{void}) \text{ } l) F \dots) FS_1 \dots))$
 where $v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{a}}$
 $(t_0 \sigma Q_0 (FS_{\text{main}} FS_0 \dots (\text{stack } FS_1 \dots)) \longrightarrow$ [dequeue-cancell
 $(t_1 \sigma Q_1 (FS_{\text{main}} FS_0 \dots (\text{stack (frame (throw-in! (tag } x_{\text{tag}} \text{ "cancelled") } x_{\text{async}})) FS_1 \dots))$
 where $(\text{frame (resume! (tag } x_{\text{tag}} \text{ } _) x_{\text{async}}) Q_1) = \text{q-pop}[Q_0], \text{task-cancelled?}[\text{lookup}[\sigma, x_{\text{async}}]], \text{all-busy?}[FS_0, \dots], t_1 = \text{step}$
 $(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{os/io natural } v)] \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [os
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } E[(\text{spawn (tag } x_{\text{tag}})]) \text{ } l) F \dots) FS_1 \dots))$
 where $(x_{\text{dummy}} \text{ } x_{\text{tag}}) = \text{lib:gensyms}[\sigma_0, \sigma_0], \sigma_1 = \text{ext1}[\sigma_0, (x_{\text{tag}} \text{ (coroutine (lambda (x_{\text{dummy}})$
 $\quad (\text{begin } x_{\text{dummy}}$
 $\quad \quad \text{lib:while}[(\leq (\text{os/time}) \text{ lib:}\Sigma[t_0, \text{natural}$
 $\quad \quad \quad (\text{yield (tag } x_{\text{tag}})])]$
 $\quad \quad \quad v)))]]$
 $(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{os/time}) \text{ } l) F \dots) FS_1 \dots)) \longrightarrow$ [os/tir
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[t_0] \text{ } l) F \dots) FS_1 \dots))$
 $(t_0 \sigma Q_0 (FS_{\text{main}} FS_0 \dots (\text{stack } FS_1 \dots)) \longrightarrow$ [dequeue
 $(t_1 \sigma Q_1 (FS_{\text{main}} FS_0 \dots (\text{stack q-head } FS_1 \dots))$
 where $\text{q-head} = (\text{frame } _ x_{\text{async}}), (\text{q-head } Q_1) = \text{q-pop}[Q_0], (\text{not task-cancelled?}[\text{lookup}[\sigma, x_{\text{async}}]]], \text{all-busy?}[FS_0, \dots], t_1 = \text{step}$
 $(t_0 \sigma Q ((\text{stack (frame } E[(\text{block (tag } x_{\text{coro}})]) \text{ } l) FS_1 \dots)) \longrightarrow$ [block
 $(t_1 \sigma Q ((\text{stack (frame } E[(\text{block (spawn (tag } x_{\text{coro}})]) \text{ } l) FS_1 \dots))$
 where $(\text{sync? } l), (\text{not any-busy?}[FS_{\text{rest}}, \dots]), (\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]], \text{none} = \text{find-unawaited-error}$
 $(t_0 \sigma () ((\text{stack (frame } E[(\text{block (task } x_{\text{async}})]) \text{ } l) FS_{\text{rest}} \dots)) \longrightarrow$ [block-e
 $(t_1 \sigma () ((\text{stack (frame } E[v] \text{ } l) FS_{\text{rest}} \dots))$
 where $(\text{sync? } l), (\text{not any-busy?}[FS_{\text{rest}}, \dots]), (\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]], \text{none} = \text{find-unawaited-error}$
 $(t_0 \sigma () ((\text{stack (frame } E[(\text{block (task } x_{\text{async}})]) \text{ } l) FS_{\text{rest}} \dots)) \longrightarrow$ [block-f
 $(t_1 \sigma () ((\text{stack (frame } E[(\text{throw } v)] \text{ } l) FS_{\text{rest}} \dots))$
 where $(\text{sync? } l), (\text{not any-busy?}[FS_{\text{rest}}, \dots]), (\text{some } v) = \text{find-unawaited-error}$